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THE OPERATING SYSTEM FOR TRAILER DEALERS

Bid & Delivery Pack

Companion ③ of three. Everything a bidder and a buyer need to price and plan the build of the **Product Specification**: the MVP scope cutline, the integration register, the assumptions & RACI, an estimation worksheet, and the stage-gated delivery plan from internal demo to GA.

Document v1.0 · **Non-normative, all figures PROPOSED** — scope tags, vendors, effort ranges and stage gates are planning positions for the client to confirm, not commitments · Companions: ① Product Specification · ② Functional Specification Proposal · Confidential draft

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1 · How to Use This Pack — the Four Levers

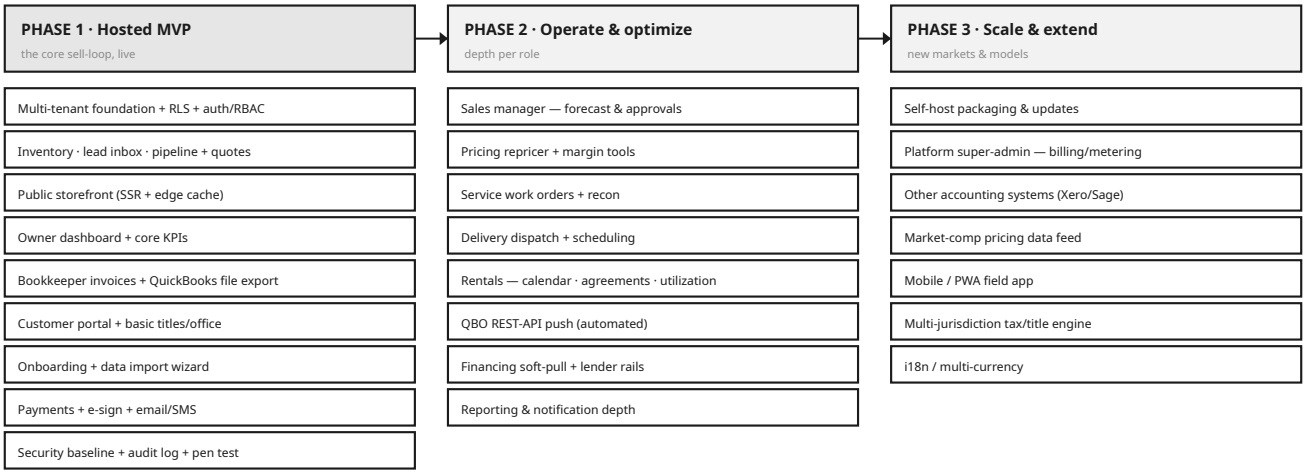
The **Product Specification** describes the full product; a bid priced against "all of it, at once" is neither costable nor advisable. This pack cuts the work into priceable shape. Four decisions — **the levers** — swing the price more than everything else combined, and they are the client's calls:

Lever	Proposed default	Bid impact if changed
Scope cutline	Phase-1 MVP only (§2)	Pricing all phases at once multiplies the bid 2–3×
Delivery model	Hosted multi-tenant only; self-host deferred	Adding self-host ≈ +30–50% (dual release/test/support)
Design ownership	Bidder produces visual design from the Spec's IA wireframes	Client-supplied hi-fi designs remove a whole phase
Launch jurisdiction	US, one-to-few states, USD, English	Multi-state tax/title + i18n are sizeable adders

Everything downstream — the MoSCoW tags (§2), the vendor register (§3), the assumptions (§4), the effort ranges (§5) and the stage gates (§6) — tightens dramatically once these four are fixed. Read with: ① the **Product Specification** (the requirements being priced) and ② the **Functional Specification Proposal** (the proposed technical solutions the estimate assumes).

2 · MVP Scope Matrix & Release Roadmap

The specification describes the **full** product. To bid accurately, the work must be cut into releases — otherwise a bidder prices "everything at once," which is neither costable nor advisable. This appendix proposes a **Phase-1 (MVP) cutline** and tags every capability MoSCoW. PROPOSAL The cutline is the client's to confirm; it is the single largest driver of the bid.



Proposed three-phase roadmap. Phase 1 delivers the core sell-loop as a hosted product; Phase 2 adds per-role depth; Phase 3 opens new markets and delivery models. Self-host, the platform-admin billing plane, additional accounting systems and a mobile field app are deliberately deferred — each is a major cost lever that should not inflate the first bid.

2.1 Roles — Phase-1 cutline

Role (spec §6)	Phase 1	Rationale
Salesperson	**MUST**	Pipeline + quote builder is the core revenue loop
Inventory Manager	**MUST**	Units are the spine; nothing sells without inventory
Lead Generation / BDC	**MUST** (lite)	A unified lead inbox; advanced source analytics → Should
Owner / Dealer Principal	**MUST** (lite)	Core dashboard + alert feed; deep drill-downs → Should
Bookkeeper / Accounting	**MUST** (lite)	Invoices/payments + **file** accounting export (Proposal §10 v1)
Customer (Buyer) Portal	**SHOULD**	Quote view + deal tracker; financing/e-sign can lag
Office / Titles & Admin	**SHOULD** (basic)	Manual title tracker; full per-state DMV engine → Could
Sales Manager	**SHOULD**	Forecast/approvals add value once a team exists

Pricing Analyst	**COULD**	Repricer depends on market-comp data (Phase 3)
Service Technician	**COULD**	Work-order board; recon can be tracked manually at MVP
Delivery Dispatch	**COULD**	Delivery can be a simple status until volume warrants
Platform Super-Admin	**WON'T** (now)	Tenant provisioning can be operator-run/manual at launch

2.2 Capabilities & cross-cutting — Phase-1 cutline

Capability	Phase 1	Note
Multi-tenant foundation + RLS + auth/RBAC	**MUST**	Non-negotiable platform floor (Spec App. B)
Public storefront (SSR + edge cache, lead forms)	**MUST**	The demand engine; SEO/first-paint matter
Onboarding + data import wizard (Spec App. C)	**MUST**	No tenant goes live without it; the activation bottleneck
Security baseline + audit log + pen test (Spec App. B)	**MUST**	Legally required once financing is offered
Notifications (transactional email/SMS)	**MUST** (core set)	Lead/quote/payment/delivery; full catalog (H) → Should
Global search	**SHOULD**	Postgres full-text at MVP; dedicated engine later
Reporting depth beyond core KPIs	**SHOULD**	Materialized core KPIs in MUST; ad-hoc/BI → Could
Financing automation + soft credit pull	**COULD**	Manual financing status at MVP; rails are heavy (§3)
Trailer rentals module (calendar, agreements, cycle billing, inspections, COI gates, utilization)	**SHOULD** (Phase 2)	A second revenue stream and a sizeable module (recurring billing + its own state machine + ASC 842 treatment) — keep out of the Phase-1 cutline unless rentals are the dealer's core business, in which case it swaps in and something else moves out
QBO REST-API accounting push	**COULD**	File export ships first (Proposal §10); API needs review
Per-jurisdiction tax/title engine (multi-state)	**COULD**	Launch in 1–2 states; generalize later
Self-host single-tenant delivery	**WON'T** (now)	~Doubles release/test/support surface — defer (§4)
Mobile / PWA field app	**WON'T** (now)	Responsive web at MVP; native field app → Phase 3
i18n / multi-currency	**WON'T** (now)	US-English / USD at launch unless required

3 · Integration & Third-Party Register

Integrations are where estimates blow up: each is a vendor choice, an integration build, a credential/account dependency, and sometimes a **multi-week production-review** gate (QuickBooks and Stripe both have one). Vendor picks below are RECOMMENDED defaults, not mandates. “Owner” = who holds the contract/keys.

Service	Recommended	Phase	MVP build vs stub	Owner	Review / lead time	Notes
Payments (card + ACH)	Stripe Connect (direct charges)	1	Build	Each dealer = merchant	Connect onboarding per dealer	Keeps each dealer at PCI SAQ A; tokenize (Spec App. B)
E-signature	Dropbox Sign / DocuSign	1–2	Build (basic)	Platform	Low	ESIGN/UETA consent flow (Spec App. B.3); not for titles
Transactional email	Postmark / SendGrid	1	Build	Platform	Domain/DKIM setup	Per-tenant from-domain & deliverability
SMS	Twilio	1	Build	Platform	10DLC registration	Delivery/appointment alerts; A2P registration lead time
Object storage	S3 (MinIO for self-host)	1	Build	Platform	None	Photos + signed docs; off-DB (ADR #6)
Cloud / hosting	One major cloud + managed Postgres	1	Build	Platform	None	Drives infra/DevOps line (§4/§5)
QuickBooks	File export → Intuit dev app	1 → 2	Build (file)	Platform	**API: multi-week review**	File needs no review; API gated (Proposal §10)
Soft credit pull / prequal	700Credit / bureau reseller	2	Stub at MVP	Dealer-contracted	Vendor onboarding + FCRA	Triggers GLBA/Red-Flags scope (Spec App. B)
Financing / lender rails	RouteOne / Dealertrack	2–3	Stub at MVP	Dealer-contracted	Heavy, per-lender	Often manual at launch; big integration if automated
Freight / lane rates	Carrier API / manual	2–3	Stub / manual	Platform	Varies	Dispatch is Phase-2+ (§2)
Market-comp pricing data	vAuto / J.D. Power feed	3	Out of MVP	Dealer-contracted	Commercial deal	Expensive licensed data; powers the repricer
Error/uptime monitoring	Sentry + metrics stack	1	Build	Platform	None	Required by NFR observability + Spec App. B logging

Bidder takeaways. Only a handful are true MVP builds (payments, email/SMS, e-sign, storage, monitoring, QuickBooks file export). The expensive, slow-onboarding ones (credit pull, lender rails, market-comp data) are **stub-at-MVP** and dealer-contracted — confirm each is out of the Phase-1 price. Two carry **production-review lead times** that affect the schedule, not just effort: **Stripe Connect** (per-dealer onboarding) and the **QuickBooks API** (deferred to Phase 2 specifically to avoid blocking MVP).

4 · Assumptions, Delivery Model & RACI

A specification is not a statement of work. These are the assumptions and ownership boundaries a bidder must fix before pricing. Defaults are (PROPOSED); each is a decision for the client.

4.1 Load-bearing assumptions (confirm each)

Open decision	Proposed default	Bid impact if changed
Release scope priced	**Phase 1 (MVP)** only (§2)	Pricing all phases at once multiplies the bid 2–3×
Delivery model at launch	**Hosted multi-tenant only**; self-host deferred	Adding self-host ≈ +30–50% (dual release/test/support)
Visual/UX design	**Bidder produces** design system + screens from the IA wireframes	If client supplies hi-fi designs, remove the design phase
Launch jurisdiction	**US, single state/handful**, USD, English	Multi-state tax/title + i18n are sizeable adders
Mobile	**Responsive web**; no native/PWA app	A field app is a separate workstream (Phase 3)
Data migration	Self-serve import (Spec App. C) + **assisted onboarding as a service line**	Per-format legacy DMS adapters are variable scope
Legal/compliance content	**Client provides** ToS, privacy, contract & tax/title legal text	Bidder builds the mechanism, not the legal copy (Spec App. B)
Pen test & security audit	**In scope**, third-party, once per release	Required by the Safeguards Rule; a real line item
Environments	dev / staging / prod + IaC + CI/CD	Fewer envs cut cost but raise release risk

4.2 RACI (R = responsible, A = accountable, C = consulted, I = informed)

Workstream	Client	Vendor (bidder)	Third-party
Scope & MVP cutline sign-off	**A**	R / C	—
UX/visual design	C / A	**R**	—
Build (platform, roles, storefront)	I	**R / A**	—
Integrations (payments, email/SMS, e-sign, QB)	C	**R**	C (vendors)

Vendor accounts & credentials	**A** (commercial)	C	R (provisioning)
Security controls & audit log	I	**R / A**	—
Penetration test	A	C	**R**
Legal/compliance content	**R / A**	I	C (counsel)
Data migration per dealer	C (data)	**R**	—
Infra / DevOps / backups	I	**R / A**	C (cloud)
UAT & acceptance sign-off	**R / A**	C	—
Production app reviews (QB / Stripe)	A	**R**	C (Intuit/Stripe)

4.3 Commercial wrapper — to be completed in the SOW

Placeholders a bid must fill (intentionally not invented here): milestones & payment schedule · team composition & rate basis (fixed-bid vs T&M) · change-control process · warranty/support period & post-launch SLA · IP ownership & source escrow · acceptance criteria per milestone · liability/indemnity (ties to Spec App. B). **Out of scope (Phase 1):** self-host packaging · platform billing/metering admin · additional accounting systems · market-comp pricing data · native mobile app · multi-state tax/title automation · i18n.

5 · Estimation Worksheet (WBS)

A feature-level work breakdown for **Phase-1 (MVP)** a bidder can price against. PLANNING RANGES Figures are order-of-magnitude **person-weeks (pw)** for a small senior team on the recommended stack (Proposal §11) — **not a quote**. Re-estimate against the confirmed cutline, the client's rates, and the assumptions in §4.

Work package	Effort (pw)	Risk driver
Foundation		
Multi-tenant core — RLS, tenant context, PgBouncer/SET LOCAL, auth, RBAC	6 – 10	High — isolation correctness; CI negative test
Data model + migrations + CRUD scaffolding (Spec App. A)	4 – 8	Low
DevOps — envs, IaC, CI/CD, observability, backup/DR	4 – 8	Medium
Design system + visual design from IA wireframes	6 – 12	Medium — depends on L.1 design ownership
Core domain (MVP roles)		
Inventory management (units, statuses, photos, lots)	4 – 7	Low
Lead capture + inbox + routing	3 – 5	Low
Sales pipeline + quote builder	5 – 8	Medium
Owner dashboard + KPIs (materialized read models)	3 – 6	Low
Office/titles (basic) + customer portal (basic)	4 – 7	Low
Bookkeeper invoices/payments + QuickBooks file export (Proposal §10 v1)	4 – 7	Medium — GL mapping + idempotency
Public storefront (SSR + edge cache, detail, lead forms)	4 – 7	Medium — SEO + responsive
Rental module — **Phase 2, NOT in the Phase-1 rollup** (availability calendar + agreements state machine + cycle billing + check-out/in inspections + COI gates + deposits + utilization KPIs + ASC 842 export buckets)	10 – 18	High — recurring billing, its own state machine, lessor accounting
Cross-cutting		
Permissions enforcement + field-level masking	2 – 4	Medium — pervasive
Notifications (core transactional set, Spec App. E)	3 – 5	Low

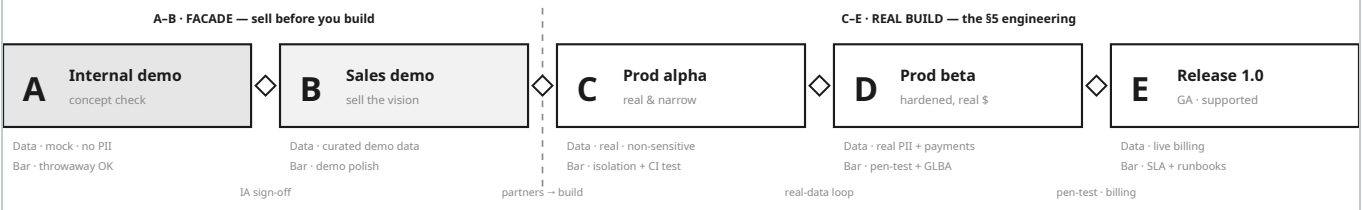
Onboarding + data import wizard (Spec App. C)	5 – 9	High — messy real-world data
Global search (Postgres FTS)	2 – 4	Low
Integrations (MVP subset, §3)		
Payments — Stripe Connect (tokenized, per-dealer)	3 – 6	Medium — Connect onboarding
E-signature + email + SMS	3 – 6	Medium — A2P/DKIM setup
Security & quality		
Security controls + audit log (Spec App. B hard gates)	4 – 7	High — compliance baseline
Third-party penetration test + remediation	2 – 4	Medium — external cost + fixes
Test automation (unit/integration/e2e + tenant-isolation)	5 – 9	Medium
UAT + hardening + bug-fix	4 – 8	Medium
Overhead		
PM / coordination / ceremonies (≈ 15–20% applied to the above)	+ overhead	—

Phase-1 rollup (engineering, excl. PM overhead): ≈ 80 – 150 person-weeks. With PM overhead, ≈ **95 – 175 pw** → roughly **6 – 12 calendar months** with a delivery team of ~3–4 FTE, plus a designer and part-time PM. **Cost** is deliberately omitted — multiply by the client's confirmed blended rate. The range is wide on purpose: it tightens dramatically once the four levers in §4 (scope cutline, delivery model, design ownership, launch jurisdiction) are fixed.

Estimation basis & caveats. Ranges assume the recommended stack (Postgres + RLS, modular monolith, batteries-included framework — Proposal §11); a senior team familiar with multi-tenant SaaS; integrations limited to the MVP subset (§3) with the rest stubbed; and the §4 assumptions holding. They exclude: self-host, the deferred integrations, the Phase-2 rental module (priced separately above), market-comp data, native mobile, multi-state tax/title, i18n, and ongoing run-cost. These are **planning figures to scope a bid**, not a fixed-price commitment.

6 · Delivery Stages (Internal Demo → GA)

A **maturity** ladder, orthogonal to the feature-depth roadmap (§2): each stage takes a slice of the Phase-1 scope and raises the bar on **what is real** — data, persistence, security, money, SLA. The decisive boundary is between **B and C**. Stages A–B are facade/prototype-grade (and A is **largely already built**); C–E are the real engineering priced in §5. The governing rule: **data sensitivity gates the quality bar, not the calendar.**



Five stage-gates from internal demo to general availability. The dashed line marks the facade→real-build boundary: A–B are a few person-weeks of polish on the existing prototype; C–E are essentially the entire §5 estimate. Tenant isolation becomes mandatory at C (real data); the legally-required pen test and GLBA Safeguards baseline land at D (real PII + money); SLA, billing and support gate E.

Stage	Audience & goal	Scope at this stage	What's real	Quality / security bar	Exit gate
A · Internal demo	Internal stakeholders — validate IA & the one-record-many-lenses idea; lock the cutline	Clickable prototype: storefront + 12 role dashboards on mock data	Nothing behind it — mock data, no auth/PII	Renders clean; throwaway code OK	IA & §2 cutline signed off · **(already built)**
B · Sales demo	Prospects, design partners, investors — sell the vision, win LOIs	Same prototype, polished: branding, stable URL, demo script, reset-to-clean data, one believable tenant	Still a facade — curated demo data, no money/PII	Demo-grade: never breaks live, phone-presentable	Partners committed · pricing validated · **go/no-go to build**
C · Production alpha	Internal + 1–2 design-partner dealers, controlled — prove the real architecture on the smallest valuable slice	Real multi-tenant core (RLS/auth/RBAC) + core MUST roles: inventory, lead, pipeline+quote, owner dashboard, storefront, import v1	**Real persistence**; real but non-sensitive data; no financing PII or money yet	Happy paths work; **tenant isolation + CI negative test green**; hardening in progress; no pen test	Sell-loop works on real data for a partner; isolation proven
D · Production beta	Beta dealer cohort with real buyers — validate under real load & money;	All MUST + most SHOULD: real payments (Stripe Connect), e-sign, QB file export, notifications, self-serve	**Real PII + real payments** — now compliance-bearing	Hardened; **pen test done + remediated**; audit log, observability,	Stable across cohort; pen-test issues closed; billing ready

	complete compliance baseline	onboarding, customer portal		backups; GLBA Safeguards baseline; SLA targeted not guaranteed	
E · Release 1.0 (GA)	Open self-serve signup, paying dealers — commercial launch of Phase 1	Full §2 Phase-1 cutline, polished; self-serve onboarding + live billing	Everything real + monetized	GA: UAT signed off, runbooks, backup/DR drills, on-call, change-control; **99.9% SLA honored**; warranty/support begins	GA criteria met → 1.0 ships; Phase-2 backlog begins

6.1 Effort & the two principles

- **A/B are facade; C/D/E are the build.** A is essentially done; B is ~1–3 pw of polish on the existing prototype. C–E together are the full **≈95–175 pw** §5 estimate, split roughly: **C** ≈ foundation + core MUST roles (~40–60% of MVP), **D** ≈ the remainder + security/QA/pen-test packages, **E** ≈ the UAT/hardening/launch-readiness tail + billing/support setup. Do not let the polished sales demo imply the system is “almost done.”
- **Data sensitivity gates the bar, not time.** Tenant isolation is required the moment real data exists (**C**). Real PII and money — and therefore the legally-required pen test and GLBA Safeguards baseline (Spec App. B) — only enter at **beta (D)**, the stage not to rush. SLA, runbooks, billing and support gate **GA (E)**.

Cross-references: stage scope maps to the MoSCoW tags and roadmap in **§2**; integration availability per stage to **§3** (e.g., QuickBooks *file* export at D, QBO API in Phase 2/post-GA; soft-credit-pull stubbed until financing); security gates to **Spec App. B**; effort to **§5**.